

MEDICINE OUTCOME PREDICTION SYSTEM

Business Report

Evidence review, model recommendation, and implementation gate

CLIENT DECISION

Proceed to controlled clinical validation, not production deployment. Use the Neural Network as the primary screening candidate and the Decision Tree as the transparent challenger.

1,050,000

cleaned records

22.3%

positive outcomes

0.410

best test F1

Executive Decision

What the analysis means for a client decision

RECOMMENDATION

The project has produced a credible analytical baseline, but not a deployment-ready clinical predictor. The Neural Network leads the held-out test set on F1 (0.410) and balanced accuracy (0.607); it is therefore the best current candidate for controlled validation. The Decision Tree is the preferred challenger because its reasoning can be inspected directly.

KEY EVIDENCE

- Outcome prevalence is 22.33%; accuracy alone rewards predicting the majority class.
- The recommended candidate recalls 64.1% of positive outcomes at threshold 0.50, with precision of 30.2%.
- XGBoost and Gradient Boosting show about 77.7% accuracy but recall only 0.7% and 0.0%; they should not be selected on accuracy.
- Age is the clearest business segmentation signal: efficacy falls from 41.23% under 18 to 13.86% at 65+.
- Model outputs are predictive associations, not treatment effects or causal recommendations.

DECISION GATES BEFORE USE

1. Lock the intended use and false-negative cost. 2. Calibrate probabilities and select a threshold on validation data. 3. Test performance by age, sex, ethnicity, drug, diagnosis, and site. 4. Run temporal and external validation. 5. Deploy only with clinician review, audit logging, monitoring, and rollback.

13-Step Delivery Summary

Complete project coverage from raw data to clinical analytics

01 Data collection and EDA

Profiled raw clinical records, distributions, missingness, duplicates, data types, and correlations.

02 Data preprocessing

Standardized placeholders, dates, dosage, duration, demographics, diagnoses, drugs, routes, and binary outcomes.

03 Feature engineering

Created age, BMI, duration, risk flags, lab abnormality count, seasonal/date features, and encoded categories.

04 Decision Tree

Interpretable baseline with class balancing and held-out evaluation.

05 Random Forest

Ensemble baseline with class balancing and held-out evaluation.

06 XGBoost

Boosted model with threshold analysis; accuracy is not a sufficient selection criterion.

07 Gradient Boosting

Boosted model with strong majority-class behavior at threshold 0.50.

08 Model comparison

Common stratified 60/20/20 split, scorecard, ROC/PR, calibration, threshold, confusion, and stability analysis.

09 Neural Network

Dense Keras classifier; highest test F1 and balanced accuracy in the saved comparison.

10 LSTM

Sequence model evaluated with sequence length 1; it does not yet exploit longitudinal sequences.

11 SHAP explainability

Global feature contribution analysis across the trained models.

12 SHAP/LIME comparison

Compared global and local explanations; agreement is descriptive and LIME artifacts require caution.

13 Clinical SQL analysis

Outcome, age, drug, adverse-event, high-risk, and dosage analyses exported to SQLite and CSV.

Data and Clinical Findings

Descriptive insights for segmentation, safety review, and prioritization

2

raw records

1

raw columns

0

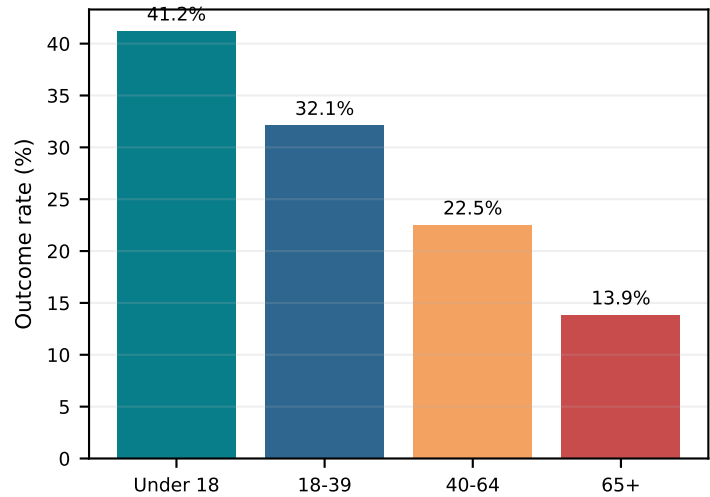
raw missing cells

0

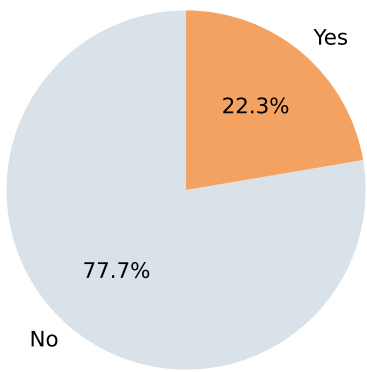
exact duplicate rows

OUTCOME AND AGE

Outcome rate declines with age



Outcome mix



BUSINESS INTERPRETATION

The outcome is imbalanced and the age gradient is large enough to support targeted review workflows. Drug efficacy rates are tightly grouped, so the data does not support declaring a single drug superior without adjustment for case mix. Dosage bands are similarly flat, which argues against dose-level conclusions from this descriptive analysis.

Finding	Observed result	Business use
Top efficacy drug	Simvastatin: 22.87%	Hypothesis for adjusted study
Lowest efficacy drug	Prednisone: 21.32%	Investigate case mix
Highest ADR cell	Ciprofloxacin, 65+: 31.11%	Prioritize safety review
High-risk cohort	70,008 records with 3+ flags	Route to clinician review
Dose range	22.29% to 22.42%	No raw dose recommendation

Analytical Method

How the project turns clinical records into a decision-support candidate

DATA CONTROLS

The project standardizes malformed text and dates, parses dosage and duration, maps clinical categories to canonical values, removes duplicate patient records by retaining the most complete row, and uses leakage-safe imputation/encoding in the modeling pipeline. Target leakage fields are excluded from model inputs.

FEATURE SET

- Demographics: age, gender, ethnicity, weight, height, BMI
- Vitals and labs: blood pressure, heart rate, temperature, hemoglobin, WBC, ALT, AST, creatinine, eGFR, HbA1c, cholesterol
- Treatment context: drug, dose, duration, route, concurrent drugs, diagnosis
- Engineered flags: obesity, underweight, hypertension, tachycardia, fever, anemia, cholesterol, diabetes, renal, polypharmacy, smoking/alcohol, lab abnormality count

EVALUATION DESIGN

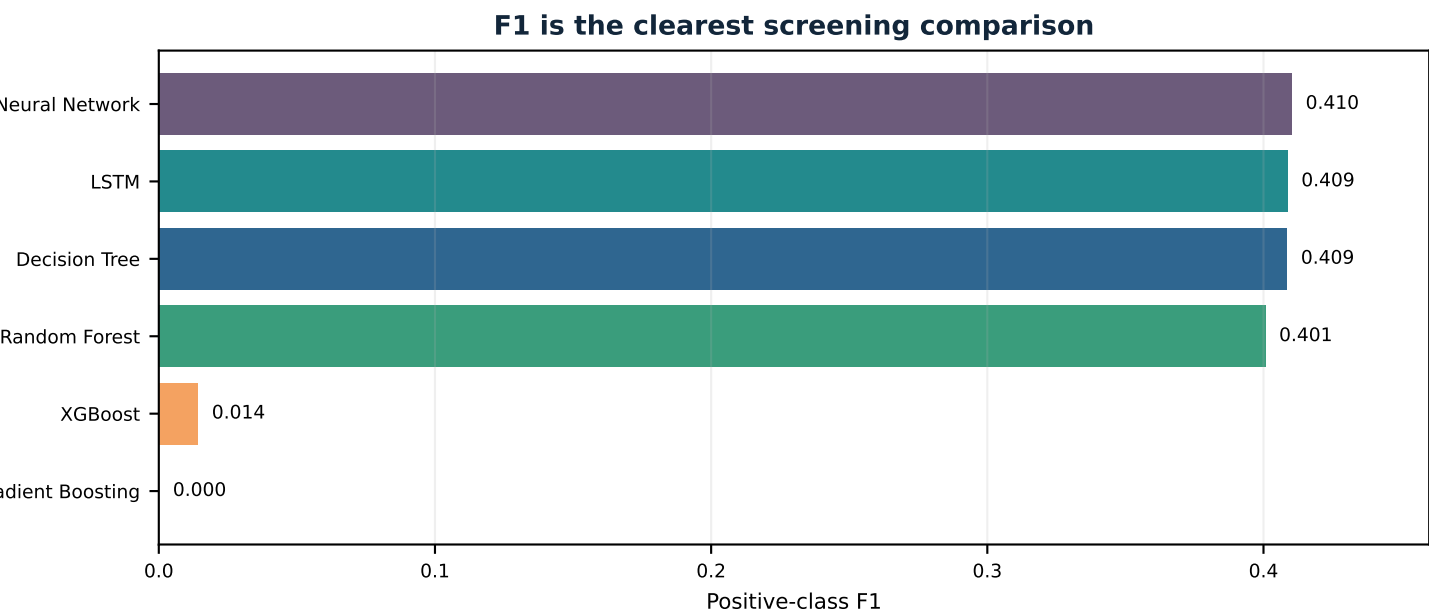
Partition	Records	Purpose
Training	591,525	Fit model parameters
Validation	197,175	Tune threshold and decisions
Testing	197,176	One-time held-out comparison
Metrics	F1, PR-AUC, recall, specificity, Br	Balance retrieval, error burden, and probability quality

The same stratified 60% / 20% / 20% split is used across models. Threshold 0.50 is shown for comparability; it must be replaced by a validated operating point tied to clinical and operational costs.

Model Scorecard

Held-out test results at threshold 0.50

Model	Acc	Bal acc	Prec	Recall	F1	PR-AUC	Brier
Decision Tree	0.591	0.606	0.302	0.632	0.409	0.326	0.235
Random Forest	0.589	0.598	0.297	0.615	0.401	0.316	0.238
XGBoost	0.777	0.503	0.508	0.007	0.014	0.331	0.166
Gradient Boosting	0.777	0.500	0.286	0.000	0.000	0.331	0.166
Neural Network	0.588	0.607	0.302	0.641	0.410	0.333	0.233
LSTM	0.599	0.607	0.305	0.620	0.409	0.331	0.232



SELECTION LOGIC

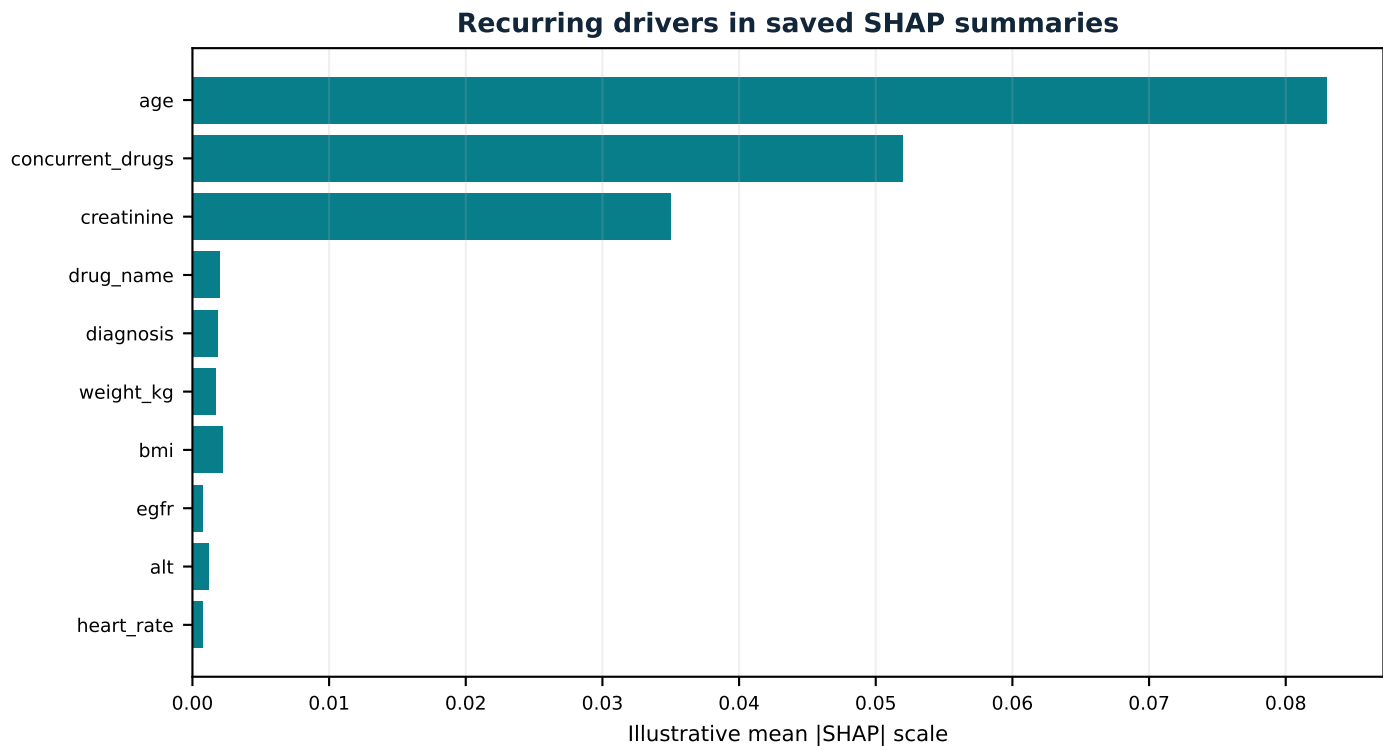
Neural Network is the current leader on F1 (0.410) and balanced accuracy (0.607); its performance is close to the Decision Tree while offering the strongest aggregate screening result. Decision Tree remains the interpretability challenger. XGBoost has the highest Brier performance among the saved models, but its default classification behavior produces too many false negatives for a positive-outcome screening workflow.

Explainability and Clinical Use

What the models appear to use, and what that does not prove

GLOBAL SHAP SIGNAL

Across the tree-based models, the most stable global contributors are age, concurrent drugs, and creatinine. Other recurring variables include drug name, diagnosis, weight/BMI, liver or renal markers, and cardiovascular measures. These are model-attribution signals, not causal effects and not prescribing instructions.



EXPLAINABILITY CONTROLS

Artifact	What it provides	Control needed
SHAP	Global contribution ranking	Review by subgroup and time
LIME	Local contribution ranking	Re-run/verify zero-valued artifacts
Decision Tree	Human-readable rules	Check stability and clinical plausibility
Model card	Intended use and limits	Maintain versioned governance record

The SHAP/LIME comparison reports limited top-10 overlap and warns that several available LIME contributions are zero. That result should be treated as an explainability QA issue, not as evidence that the corresponding models have no influential features.

Client Action Plan

A practical path from analysis to governed validation

PHASE 1 | REFRAME

Define whether the product is screening, monitoring, or retrospective analytics. Confirm the positive outcome definition, acceptable false-negative rate, review capacity, and whether the model supports care rather than automates a treatment decision.

PHASE 2 | STRENGTHEN EVIDENCE

Tune and calibrate the Neural Network and Decision Tree using validation data only. Add confidence intervals for all priority metrics, temporal holdout, external-site testing, subgroup performance, missingness sensitivity, and leakage review. Re-run LIME with validated local explanations.

PHASE 3 | PILOT WITH SAFEGUARDS

Expose risk scores and explanation summaries to trained clinicians as a second opinion. Log inputs, model version, threshold, output, human action, and outcome. Monitor calibration, drift, subgroup gaps, overrides, and false negatives; define rollback ownership before launch.

FINAL RECOMMENDATION

PRIMARY CANDIDATE

Neural Network

Best saved test F1: 0.410 | Recall: 64.1% | PR-AUC: 0.333

CHALLENGER

Decision Tree

Use for transparent review and as a benchmark for whether extra model complexity is justified.

This report summarizes observational project outputs. It does not establish treatment efficacy, safety, or causality.